



do you remember we talked about the algorithmic trading project?

Yes! We've had several conversations about your DTU Fagprojekt on day trading as a multilevel decision problem. Here's a quick recap of the key topics we covered:

- **Project framing** — Modeling day trading as a hierarchical decision system (strategic: asset selection; tactical: entry/exit)^[1]
- **Model choice** — Starting with a CNN as the foundational model, with RNN/LSTM as a potential phase-two extension^[2]
- **Pipeline setup** — Data fetching (OHLCV), CUSUM filtering, Triple Barrier labeling, normalization, and sequence creation^[3]
- **Triple Barrier method** — Every bar gets a setup, but the label depends on which barrier (profit/stop/time) is hit first^[4]
- **Adaptive sampling** — Your supervisor asked you to research sampling more frequently closer to a trade, which ties into information-driven bars like CUSUM^[5]
- **Next bar method** — Also researched as a simpler directional labeling alternative

You preferred deep learning over reinforcement learning for the project. What would you like to pick up on?

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im about the write the section about the different ML models we are testing. im starting with writing about the resnet. Do you recommend me reading this original paper?

Short answer: you don't need to read it cover to cover, but a quick skim of Sections 1–3 is worth it — especially since you're writing about it academically.

Here's what's actually useful for your section vs. what you can skip:

Worth Reading (15–20 min)

- **Abstract + Introduction** — Explains the core problem (degradation) and the motivation for residual connections. Good for framing ResNet in your own words^[6]
- **Section 3 (Deep Residual Learning)** — Covers the residual block formula $y = F(x, \{W_i\}) + x$, identity shortcuts, and bottleneck design. This is the core theory you'll need to cite^[6]
- **Figure 2 & 5** — The residual block diagrams are the canonical visuals if you need to include a figure^[6]

Can Skip

- **Section 4 (Experiments)** — Heavy ImageNet/CIFAR benchmarks. Only skim Table 3 if you want specific accuracy numbers to cite^[6]
- **Appendices A–C** — Object detection and localization results, irrelevant for a trading project^[6]

Key Points to Include in Your Section

For your Fagprojekt writeup, the essential ResNet ideas are:

- **Problem it solves:** Deeper plain CNNs suffer from a degradation problem — adding more layers increases *training* error, not just test error^[6]
- **Solution:** Instead of learning $H(x)$ directly, layers learn the residual $F(x) = H(x) - x$, making identity mapping the trivial solution^[6]
- **Skip connections:** Parameter-free identity shortcuts add no computational cost but allow gradients to flow directly through the network^[6]
- **Why it matters for you:** ResNet lets you go deeper without optimization issues, which is beneficial when extracting hierarchical temporal patterns from financial time-series with a 1D CNN adaptation^[6]

The paper is by He et al. (2015), titled "*Deep Residual Learning for Image Recognition*" — cite it as your primary source.



are resnets by nature lstm's?

No, ResNets and LSTMs are completely different architectures — they solve similar *symptoms* (vanishing gradients) but in entirely different ways.

The Key Difference

	ResNet	LSTM
Type	Feedforward CNN	Recurrent Neural Network
Processes	Spatial/structural patterns	Sequential/temporal data
Gradient fix	Skip connections (identity shortcuts)	Gating mechanisms (forget/input/output gates)
Memory	None — no hidden state	Explicit cell state across time steps
Parallelizable	Yes	No (sequential by nature)

Why They're Often Confused

Both architectures were motivated by the vanishing gradient problem. ResNet's skip connections let gradients flow directly backward through layers, while LSTM's gates control how gradients flow across *time steps*. Same problem, fundamentally different solutions.^{[7] [8]}

For Your Project

Since you're applying ResNet to financial time-series (a sequential task), it's important to acknowledge this distinction in your writeup. A 1D ResNet can extract local patterns from a fixed-length window, but it has **no memory** — it can't carry information from one bar to the next the way an LSTM can. That's exactly why combining both (or comparing them) makes for a strong multilevel decision framing in your Fagprojekt.^[9]

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You can combine them and create a ResNLS

Yes — you can combine them, but it would usually be called a ResNet-LSTM hybrid, not “ResNLS.”^{[16] [17]}

A ResNet-LSTM model uses ResNet to extract features and then feeds those features into an LSTM to model temporal dependencies across time steps or ordered inputs.^{[18] [19] [16]}

That kind of hybrid is already used in practice for sequential tasks, including prediction settings where ResNet handles representation learning and LSTM handles sequence memory.^{[20] [21] [16]}

For your project, the clean way to describe it is: “ResNet captures local patterns in each window, while LSTM captures dependencies across windows.”^{[19] [21] [16]}

If you want, I can help you write a short thesis-style paragraph introducing ResNet-LSTM for your model section.

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what is the LSTM's role in this combination of models?

The LSTM's role is **sequence memory**: it takes the features extracted by the ResNet and learns how those features evolve over time. [\[31\]](#) [\[32\]](#)

In the hybrid

ResNet usually acts as the feature extractor, turning each input window or frame into a compact representation. [\[33\]](#) [\[31\]](#)

The LSTM then processes the ordered sequence of those representations to capture temporal dependencies, long-term context, and how earlier patterns affect later ones. [\[32\]](#) [\[34\]](#) [\[31\]](#)

For your project

In a trading setup, you can think of it like this: ResNet detects local patterns inside each price window, while the LSTM learns the progression between windows. [\[35\]](#) [\[36\]](#)

So if you combine them, the LSTM is there because ResNet alone does not maintain an internal memory of past steps. [\[36\]](#) [\[31\]](#)

A thesis-friendly one-liner would be: “The LSTM module was included to model temporal dependencies in the sequence of ResNet-extracted features.” [\[31\]](#) [\[32\]](#)

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Research literature on this subject and find out how people are solving this problem: If the upper and lower barrier is hit on the same bar

Mostly, people do not solve this inside the original triple-barrier definition itself. The standard formulation says to label by the *first barrier touched*, but common López de Prado-style implementations such as `mlfinpy` work from a close-price series, which sidesteps same-bar upper/lower collisions rather than explicitly resolving them.

What literature says

The triple-barrier method is defined by the time of first touch among the profit-taking, stop-loss, and vertical barriers.

In the material I checked, that definition does not add a special OHLC tie-break rule for the case where both horizontal barriers fall inside the same candle.

How people handle it

When you only have OHLC data for a bar, the intrabar order cannot be determined from that bar alone.

Because of that, practitioners usually handle the problem in one of two ways: use more granular data to reconstruct the path, or apply a deterministic priority rule when the case is ambiguous. WealthLab explicitly discusses priority choices such as Favor Stop, Favor Limit, or Neutral for truly ambiguous same-bar cases, and also points to granular processing as the best way to reduce ambiguity.

Best practice for you

For your project, the most defensible rule is:

- Use lower-timeframe data for labeling whenever possible.
- If both barriers are crossed in the same bar and no finer data exists, mark the event as ambiguous.
- Resolve ambiguity with one fixed rule and state it clearly in the methodology.

If you want a conservative rule, use “favor the adverse barrier” rather than the profitable one. If you want, I can turn this into a thesis-ready methodology paragraph with formal wording.

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